BALWYN HIGH SCHOOL

Student Name	ANSWERS				
Teacher	AHU	COZ	GSL	HAL	MDI

CHEMISTRY
Unit 3, SAC 2
Test on concepts from Chapters 7, 8 and 9 of Heinemann Chemistry 2
Rate and extent of chemical reactions, and electrolysis

31 May 2021, Period 4

Reading time: 5 minutes
Writing time: 65 minutes

Structure of book

Task and Section	Number of questions	Number of questions to be answered	Number of marks
Section A Multiple Choice Questions	10	10	10
Section B Short Answer Questions	5	5	34
			Total 44

- Students are permitted to bring into the assessment room: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.
- Students are not permitted to bring into the examination room: blank sheets of paper and/or white out liquid/tape.

Materials supplied
- This question and answer book of **12 pages**.
- A data book

Instructions
- Write your name in the space provided above on this page.
- All responses must be written in English.
- Working out must be shown for all calculations
- Do NOT write on the data book

At the end of the assessment session
- Wait silently for the supervisor to collect your **booklet and data book.**

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the examination room.

Students must not disclose the contents of the task; to do so will be a breach of VCE guidelines and will be dealt with according to VCAA regulations.

Section A – Multiple Choice Questions

Instructions for Section A

Answer **all** questions in pencil on the answer sheet provided for multiple choice questions.
Choose the response that is **correct** or that **best answers** the question.
A correct answer scores 1, an incorrect answer scores 0.
Marks will **not** be deducted for incorrect answers.
No marks will be given if more than one answer is completed for any question.

Question 1

Which of the following explains why decreasing the concentration of the reactants in an aqueous reaction will lower the rate of reaction?

A. There will be fewer particles present.

B. The average kinetic energy of the particles present in the solution will be lower.

C. There will be a lower number of particles per volume with sufficient energy for a fruitful reaction to occur.

D. There will be a decrease in the frequency of fruitful collisions between the reacting particles.

Question 2

Increasing pressure of a gas mixture in a chemical reaction will increase the reaction rate by

A. increasing the frequency of collisions

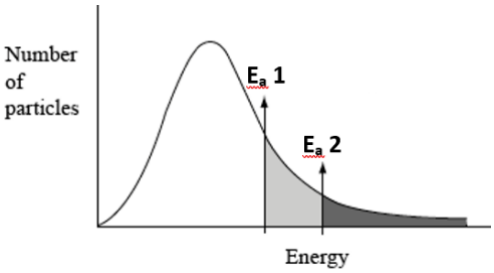
B. lowering the activation energy

C. increasing the average kinetic energy of the reactant particles

D. increasing the proportion of successful collisions

Question 3

Consider the following Maxwell-Boltzmann distribution graph below.



E_a 1 represents

A. The proportion of successful collisions with enough energy before the temperature was increased.

B. The proportion of successful collisions with enough energy after the temperature was increased.

C. The activation energy for the reaction before a catalyst was added.

D. The activation energy for the reaction after a catalyst was added.

Use the following information to answer Questions 4, 5 and 6

Methanol, CH_3OH, can be used as a fuel. The reaction for the commercial production of methanol can be represented by the following equation.

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g) \qquad \Delta H = -90 \text{ kJ mol}^{-1}$$

A gaseous mixture of CO, H_2 and CH_3OH is at equilibrium at 300°C.

Question 4

What is the effect on the mass of CO and H_2 at equilibrium when more CO is added at constant volume and temperature and a new equilibrium is established?

	Effect on **m(CO)** at equilibrium	Effect on **m(H_2)** at equilibrium
A.	decrease	increase
B.	increase	decrease
C.	increase	increase
D.	decrease	decrease

Question 5

What is the effect on the concentration of CO and CH_3OH at equilibrium the **instant** the volume of the vessel is increased at constant temperature?

	Effect on **[CO]**	Effect on **[CH_3OH]**
A.	decrease	increase
B.	increase	decrease
C.	increase	increase
D.	decrease	decrease

Question 6

The expression for the equilibrium constant for the reverse reaction is:

A. $\frac{[CO][H_2]}{[CH_3OH]}$

B. $\frac{[CO][H_2]^2}{[CH_3OH]}$

C. $\frac{[CH_3OH]}{[CO][H_2]}$

D. $\frac{[CH_3OH]}{[CO][H_2]^2}$

Question 7

Determine the value of the equilibrium constant (K_c) for the following hypothetical reaction, given that at equilibrium, [A] = 0.001 mol L^{-1}, [B] = 0.1 mol L^{-1} and [C] = 0.2 mol L^{-1}.

$$A(g) \rightleftharpoons 2B(g) + C(g)$$

A. 0.5 M^{-2}

B. 0.5 M

C. 2 M^2

D. 20 M

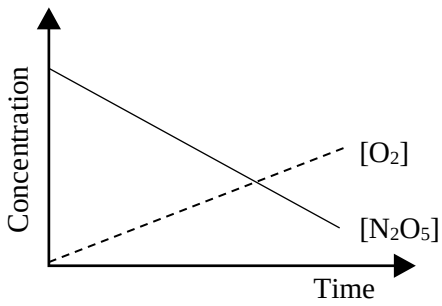
Question 8

The thermal decomposition of nitrogen pentoxide can be described by the following equation:

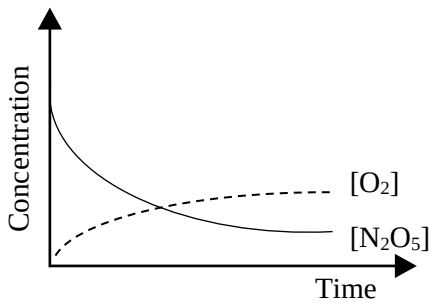
$$2N_2O_5\ (g) \rightleftharpoons 2N_2O_4\ (g) + O_2\ (g)$$

Which of the following graphs would best represent how the concentration of nitrogen pentoxide and oxygen change during the course of a reaction?

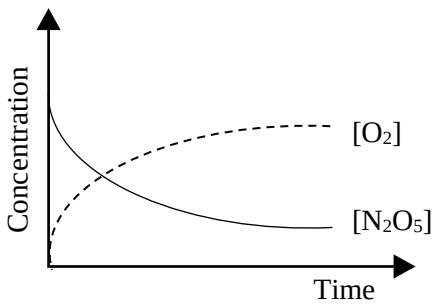
A.



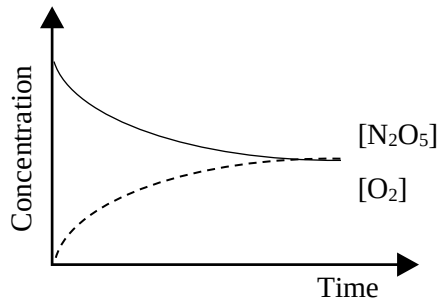
B.



C.



D.



Question 9

Which of the following experiments would not result in the production of metallic iron, Fe?

A. Add silver pieces to a solution containing iron(II) ions

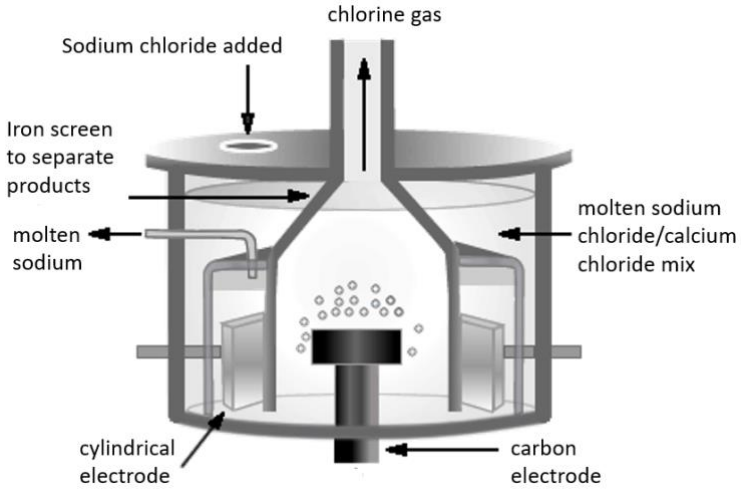
B. Add aluminium pieces to a solution containing iron(II) ions

C. Electrolysis of an aqueous solution of 1M iron(II) chloride

D. Electrolysis of molten iron(II) chloride

Question 10

A simplified diagram of the Down's cell is shown below. This cell is designed to produce molten sodium and chlorine gas in commercial quantities.



Which of the following correctly shows both the equation for the reaction occurring at the anode and the polarity of the carbon electrode?

	Reaction occurring at the anode	Polarity of the carbon electrode
A.	$Na\ (l) \rightarrow Na^+\ (l) + e^-$	negative
B.	$2Cl^-\ (l) \rightarrow Cl_2\ (g) + 2e^-$	negative
C.	$Na\ (l) \rightarrow Na^+\ (l) + e^-$	positive
D.	$2Cl^-\ (l) \rightarrow Cl_2\ (g) + 2e^-$	positive

END OF SECTION A

Multiple Choice Answer Sheet

This answer sheet must not be detached from this booklet.

INSTRUCTIONS FOR MULTIPLE CHOICE ANSWER SHEET

Use a **PENCIL** for **ALL** entries. For each question, shade the box which indicates your answer.
Marks will **NOT** be deducted for incorrect answers.
NO MARK will be given if more than **one** answer is completed for any question.
If you make a mistake, **ERASE** the incorrect answer - **DO NOT** cross it out.

1	A	B	C	**D**
2	**A**	B	C	D
3	A	B	C	**D**
4	A	**B**	C	D
5	A	B	C	**D**
6	A	**B**	C	D
7	A	B	**C**	D
8	A	**B**	C	D
9	**A**	B	C	D
10	A	B	C	**D**

Section B – Short Answer Questions

Instructions for Section B

Answer **all** questions in the spaces provided.
To obtain full marks for your responses you should:

- Give simplified answers with an appropriate number of significant figures to all numerical questions; unsimplified answers will not be given full marks.
- Show all working in your answers to numerical questions. No credit will be given for an incorrect answer unless it is accompanied by details of the working.
- make sure chemical equations are balanced and that the formulas for individual substances include an indication of state; for example, $H_2(g)$; NaCl(s)

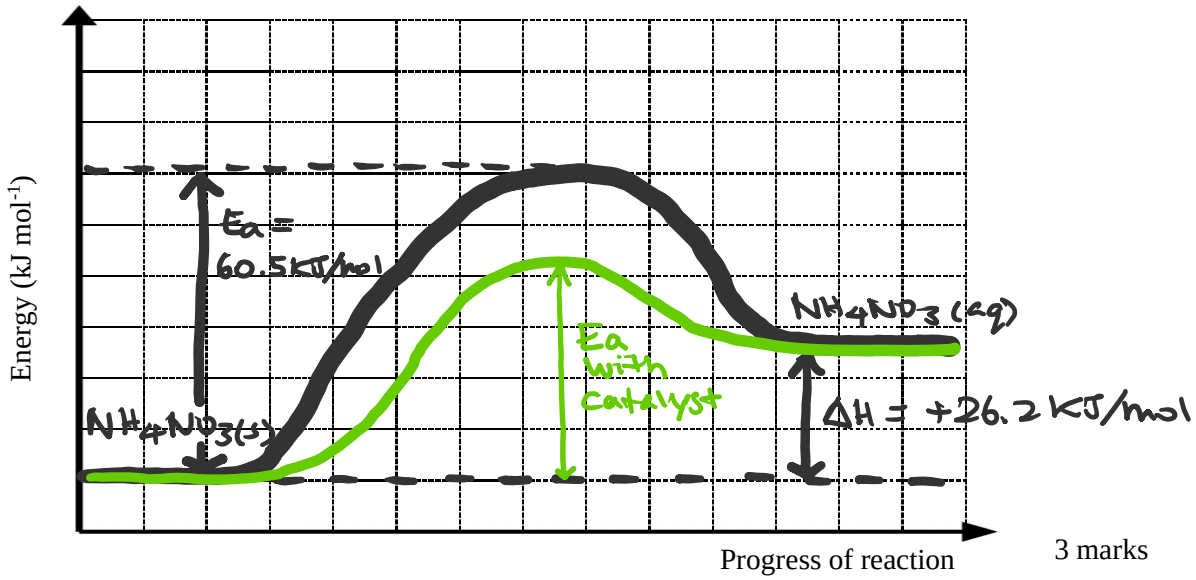
Question 1 (7 marks)

An 'instant cold' ice pack contains a bag of water inside a larger bag of finely powdered ammonium nitrate, NH_4NO_3. When the pack is squeezed, the bag of water breaks and the NH_4NO_3 dissolves as shown in the following equation:

$$NH_4NO_3(s) \rightleftharpoons NH_4NO_3(aq) \qquad \Delta H = +26.2 \text{ kJ mol}^{-1}$$

a. The activation energy for the forwards equation is 60.5 kJ mol^{-1}.

i. Draw an energy profile diagram showing the energy changes as the powder dissolves.



3 marks

1m – shape (endothermic) including reactants and products with states
1m - Ea and Delta H correct arrows and labels
1m – Ea and Delta H values included with correct units

ii. Draw a second line on your energy profile above, to show the change that would result from the use of a catalyst in this reaction. Clearly label this new line.

1 mark

b. With reference to collision theory, state and explain what would happen to the rate of the forward reaction if a solid block of the same mass of NH_4NO_3 were used instead of the fine powder.

A solid block has a **smaller surface area** than the powder and/or there are **fewer exposed particles** that are able to collide. (1 mark)

Reaction rate would **decrease** (1 mark)

This decreases the frequency of collisions as well as the **frequency of successful collisions** (1 mark)
Note: frequency is different to:

- chance/probability/percentage. The only thing that affects chance/probability/percentage is temperature.
- Number. Number does not make reference to the 'time'. You could have written **number** of collisions per **time unit (frequency!).**

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3 marks

Question 2 (5 marks)

Haemoglobin is responsible for transporting oxygen around the body. Carbon monoxide molecules can bind to haemoglobin. Carbon monoxide poisoning occurs even when there is a low build-up of carbon monoxide (CO) in the bloodstream.

Two equations relevant to these interactions are shown below, where Hb_4 represents haemoglobin.

I $Hb_4 + O_2 \rightleftharpoons Hb_4O_2$ K_1

II $Hb_4 + CO \rightleftharpoons Hb_4CO$ K_2 , where $K_2 > K_1$

a. Referring to the equations above, explain how the build-up of carbon monoxide leads to a shortage of available oxygen in body tissue and organs.

As **K_2 is greater than K_1**, and one of:

- **forwards reaction of equation II is more likely to occur** than that of equation I when CO and O_2 are both in the system
- **CO will react with haemoglobin instead of O_2**
- **Hb_4CO will be produced instead of Hb_4O_2**
  (1 mark)

And one of:

- Production of Hb_4CO reduces concentration of Hb_4, **shifting equilibrium of equation I backwards, reducing Hb_4O_2 and oxygen supply** to different parts of the body
- Because Hb_4O_2 is not being produced**, O_2 cannot be transported** around the body.
- Because Hb_4CO is being produced**, CO is transported** around the body **instead of O_2**.
  (1 mark)

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2 marks

b. Using Le Châtelier's principle explain how the administration of large doses of oxygen, O_2, could affect equation I and then equation II, in order to be used as a method to treat carbon monoxide poisoning effectively.

According to LCP, if O_2 (reactant), is added to the bloodstream, the system will shift to **partially oppose the change** and thus reduce the concentration of O_2. (1mark).

This will result in a net **forwards reaction/shift to the right in equilibrium I**. (1 mark)

As a result there will be a **decrease in concentration of Hb_4** that **leads to a** net **backwards reaction/shift to the left in equilibrium II** (1mark).

The concentration of Hb_4CO will be reduced, Hb_4 will be available to react with O_2 and Hb_4O_2 can transport O_2 around the body.

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3 marks

Question 3 (5 marks)

0.0200 mol PCl_5 was placed in a 2.00L flask at 523 K. It partially dissociates according to the equation:

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

At equilibrium, the concentration of chlorine was 0.00870 M.

a. Write an equilibrium expression for this reaction.

$$K_c = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$

1 mark

b. Determine the equilibrium concentrations of PCl_5 and PCl_3 and calculate the equilibrium constant for the reaction.

RICE table

R	PCl_5	PCl_3	Cl_2
I	0.0200 mol	0 mol	0 mol
C	Decrease by 0.0174 mol or 0.00870 M	Increase by 0.0174 mol or 0.00870 M 0.0174 mol	Increase by 0.0174 mol or 0.00870 M *$n(Cl_2)$ = 0.00870 × 2 = 0.0174mol*
E	*$n(PCl_5)$*= 0.0200 – 0.0174 = **0.0026 mol** or $[PCl_5] = \frac{0.0026}{2}$ = **0.0013M** (1 mark)	*$n(PCl_3)$*= **0.0174 mol** or $[PCl_3]$ = **0.00870M** (1 mark)	$[Cl_2]$ = 0.00870M

$$K_c = \frac{(0.00870)(0.00870)}{(0.0013)} = \mathbf{0.0582M}$$

final answer with correct units and significant figures, consequential to equilibrium values and expression from 3a (1 mark).

3 marks

c. Referring to the equilibrium constant calculated above, comment on the extent of the reaction.

Consequential to *part (b)*

K value is between 10^{-4} and 10^{4}. Therefore, extent is significant. Neither side is favoured and there are **appreciable concentrations of reactants and products** when equilibrium is established

Note: there is **not** an equal amount of reactants and products in this equilibrium.

Possible incorrect, but consequential answers:

- K >10^4 Extent is high
  The forwards reaction has been favoured (or almost complete reaction has occurred before equilibrium was established
- K <10^{-4}

Extent is very low. Negligible forwards reaction has occurred or there is very little product formed when equilibrium is established

1 mark

Question 4 (7 marks)

At 257 °C, the equilibrium constant for the following reaction is 100 M^{-2}.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

a. A 2.0 L vessel at 257 °C was found to contain 0.1 mol N_2, 0.2 mol H_2 and 0.5 mol NH_3.

i. Use expressions to show that this reaction has not yet reached equilibrium.

$[N_2]$ = 0.1/2 = 0.05 M **$[H_2]$** = 0.2/2 = 0.1 M **$[NH_3]$** = 0.5/2 = 0.25 M (1 mark)

$$Q = \frac{[\mathbf{NH_3}]^2}{[\mathbf{N_2}][\mathbf{H_2}]^3} = \frac{(0.25)^2}{0.05 \times (0.1)^3} = 1250\ \mathrm{M}^{-2}$$ (1m for correct Q expression)

(mark not awarded if referred to as K rather than Q).

When a system is at equilibrium Q = K. This **Q value** (1250) is **higher than/not equal to K** (100), therefore this reaction is **not at equilibrium** (1 mark)

3 marks

ii. In which direction does this reaction need to proceed in order to reach equilibrium?

1250 M^{-2} > 100 M^{-2} (given K value); a **net reverse reaction/shift to the left** needs to occur (1 mark)

1 mark

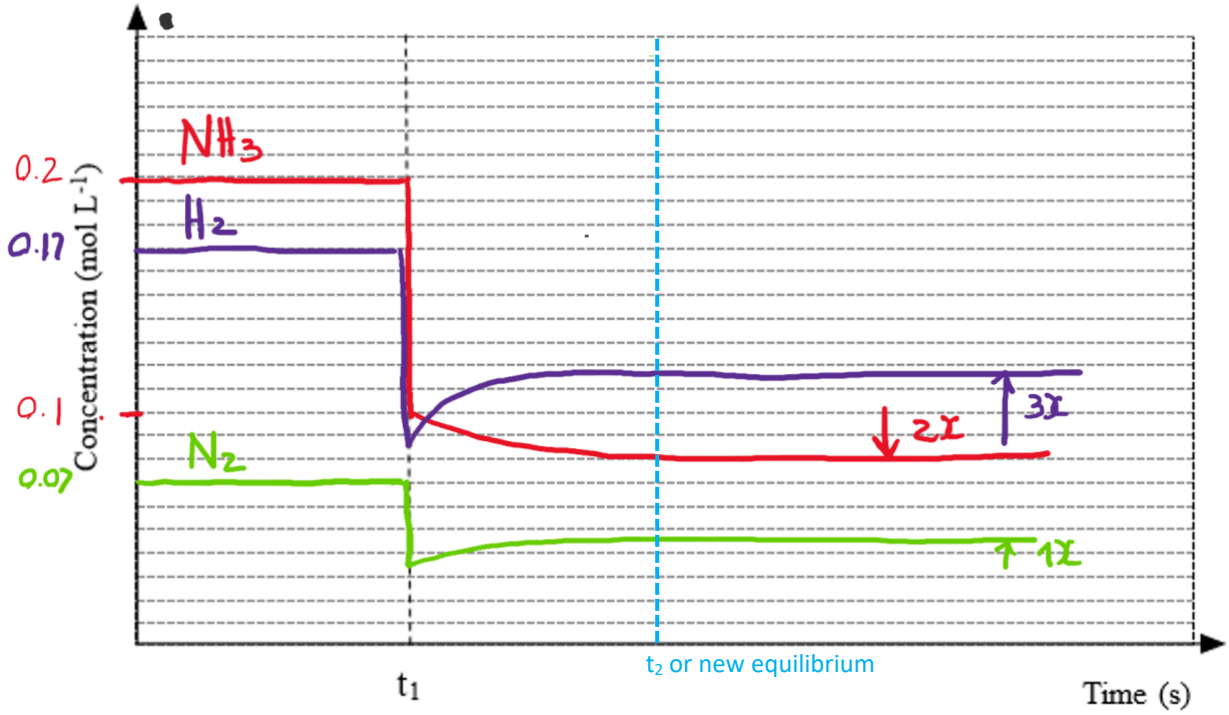
b. When the reaction had reached equilibrium, concentrations of all species were measured and plotted on the concentration time graph below.

i. On the graph, show what would happen to each species if the volume of the vessel was increased to 4.0 L at time t_1, and a new equilibrium was established.

- Correct **initial proportional drop** (**half** the initial magnitude drawn on the graph. Values not required)(1 mark)
- Correct changes in the **right direction** and **stoichiometric ratio** displayed in graph (1 mark)

ii. Indicate on the graph the point at which the new equilibrium was established.

2 + 1 = 3 marks


NH3
H2
N2
0.2
0.17
0.1
0.07
Concentration (mol L-1)
2x
3x
1x
t1
t2 or new equilibrium
Time (s)

Question 5 (10 marks)
A group of students set up two experiments to simulate the industrial extraction of nickel metal from an aqueous solution of nickel (II) ions, and solid calcium from a solution of calcium (II) ions by electrolysis.

Experiment 1	200 mL of 1.00 M $NiCl_2$ solution was electrolysed at 25ºC using inert carbon electrodes.
Experiment 2	200 mL of 1.00 M $CaCl_2$ solution was electrolysed at 25ºC using inert carbon electrodes.

a. Write the oxidation half-equation for the reaction that occurs during **Experiment 1**.
$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$ single arrow only (1 mark)

- **must** include states, **single** arrow

1 mark

b. In **Experiment 1**, a grey deposit of nickel metal formed on the cathode. What voltage could the students have used in order to achieve this result?

1.23 – (-0.25) = 1.48V, therefore **greater than 1.48V** (1 mark consequential to 5a)

Note: no mark was awarded for a single 'random' value even if higher than 1.48V. Students should be aware that calculations are required for this type of question.

1 mark

c. Assume that the students applied a steady current of 29.7 amps for 11 minutes. What mass of nickel would have deposited onto the cathode in this time?

Q (1 mark)	Q = 29.7 × (11 × 60) = 19602 C
$n(e^-)$ (1 mark)	$n(e^-) = \frac{19602}{96500} = 0.2031$ mol
n(Ni) (1 mark)	n(Ni) = ½ × 0.203 = 0.1016 mol
m(Ni) (1 mark)	m(Ni) = 0.1016 × 58.7 = 5.96g = 6.0g

4 marks

d. In **Experiment 2**, bubbles of gas appeared at both the anode and the cathode, and no calcium metal obtained. With the use of equations, explain the observations that were made in **Experiment 2**.

Reduction half-equation showing production of H_2 gas at cathode

- $2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$

Oxidation half equation showing production of O_2 gas at anode

- $2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$

(1 mark – both half equations correct, no states needed)

The bubbles observed at the **cathode are hydrogen gas**, and the bubbles observed at the **anode are oxygen gas** (1 mark)

Reference to **H_2O being the stronger oxidant than Ca^{2+} /calcium ions and stronger reductant than Cl-** /chlor**ide** ions (1 mark)

3 marks

e. How would it be possible to obtain calcium metal from calcium chloride?

Electrolysis **of molten** or liquid **calcium chloride** (1 mark)

Note: mark was awarded even if you did not mention the method used to produce calcium (e.g. electrolysis). However, this was an exception and you are not likely to get a mark for incomplete or vague answers in the future.

1 mark

END OF QUESTION AND ANSWER BOOKLET